# Unit 6, Lesson 6: Solving Real-World Problems Using Similar Triangles

## Benchmark

**MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.**

## Learning Target

☐ I can solve real-world problems using similar triangles.

## 6.6.1 Warm-Up: Brains Think About Math Visually

1. Explain a time in math class when you used a visual representation of a problem to help solve it.

2. How did that visual representation help you?

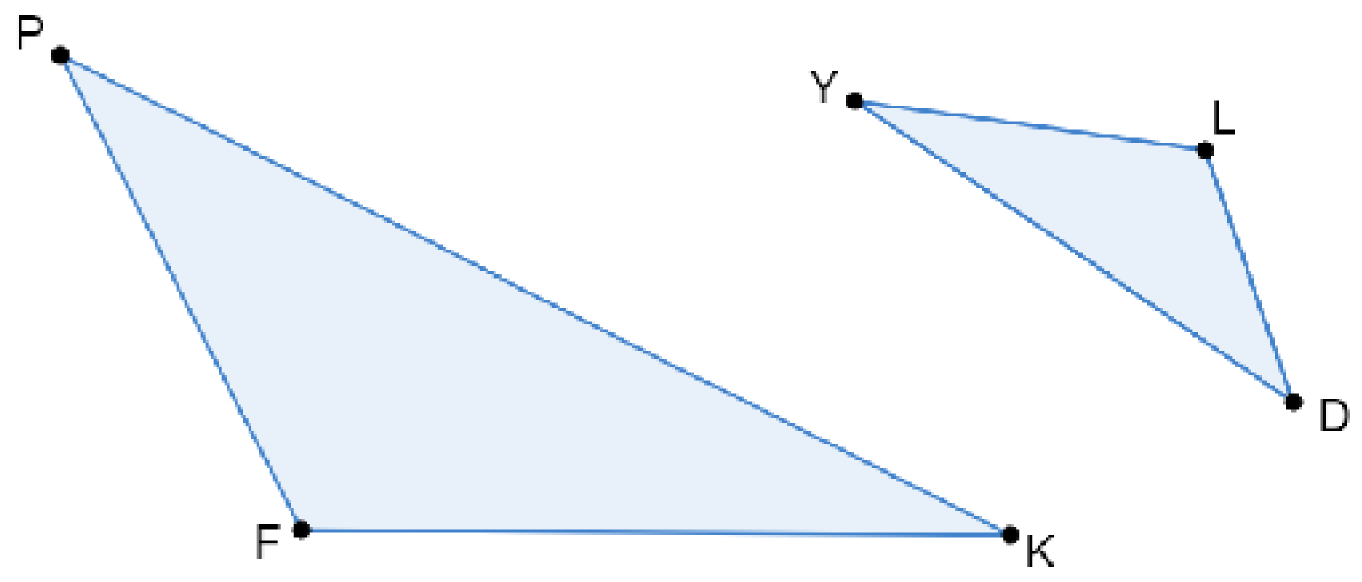





## 6.6.2 Exploration: Angles in Similar Triangles

The previous lesson introduced similar triangles and the relationship between their corresponding sides. The relative side lengths of the triangles in those situations were used to identify corresponding sides. In some cases, it may be more helpful to use angle measures to help identify corresponding sides.

Triangle $PFK$ and triangle $DLY$ are similar.




Line segment $PK$ corresponds to line segment $DY$.
Line segment $PF$ corresponds to line segment $DL$.
Line segment $FK$ corresponds to line segment $LY$.

1. Mark the figures to show which sides correspond, based on the given information, using colors or symbols.

2. Identify the angles opposite each line segment.

| Triangle $PFK$ | |
|---|---|
| Line Segment | Opposite Angle |
| $\overline{FK}$ | |
| $\overline{KP}$ | |
| $\overline{PF}$ | |

| Triangle $DLY$ | |
|---|---|
| Line Segment | Opposite Angle |
| $\overline{LY}$ | |
| $\overline{YD}$ | |
| $\overline{DL}$ | |

3. Angles opposite corresponding sides are called **corresponding angles**. Use colors or markings to show which angles correspond to each other on the triangles.

4. Complete the statement by identifying corresponding angle pairs in triangles $PFK$ and $DLY$.

$\angle PFK$ corresponds with $\angle$__________.

$\angle FKP$ corresponds with $\angle$__________.

$\angle KPF$ corresponds with $\angle$__________.

5. Consider the size of each pair of corresponding angles. With your partner, write a hypothesis regarding the relationship between each angle pair.

6. Trace a copy of triangle $PFK$ on patty paper. Be sure to label each vertex.






7. Place the copy of triangle $PFK$ on top of triangle $DLY$ so that angle $PFK$ overlaps angle $DLY$. What do you notice?

8. Repeat the process for each pair of corresponding angles.

9. Discuss with your partner whether your observations support or disprove your hypothesis about the relationship between each angle pair. Then, summarize your discussion.

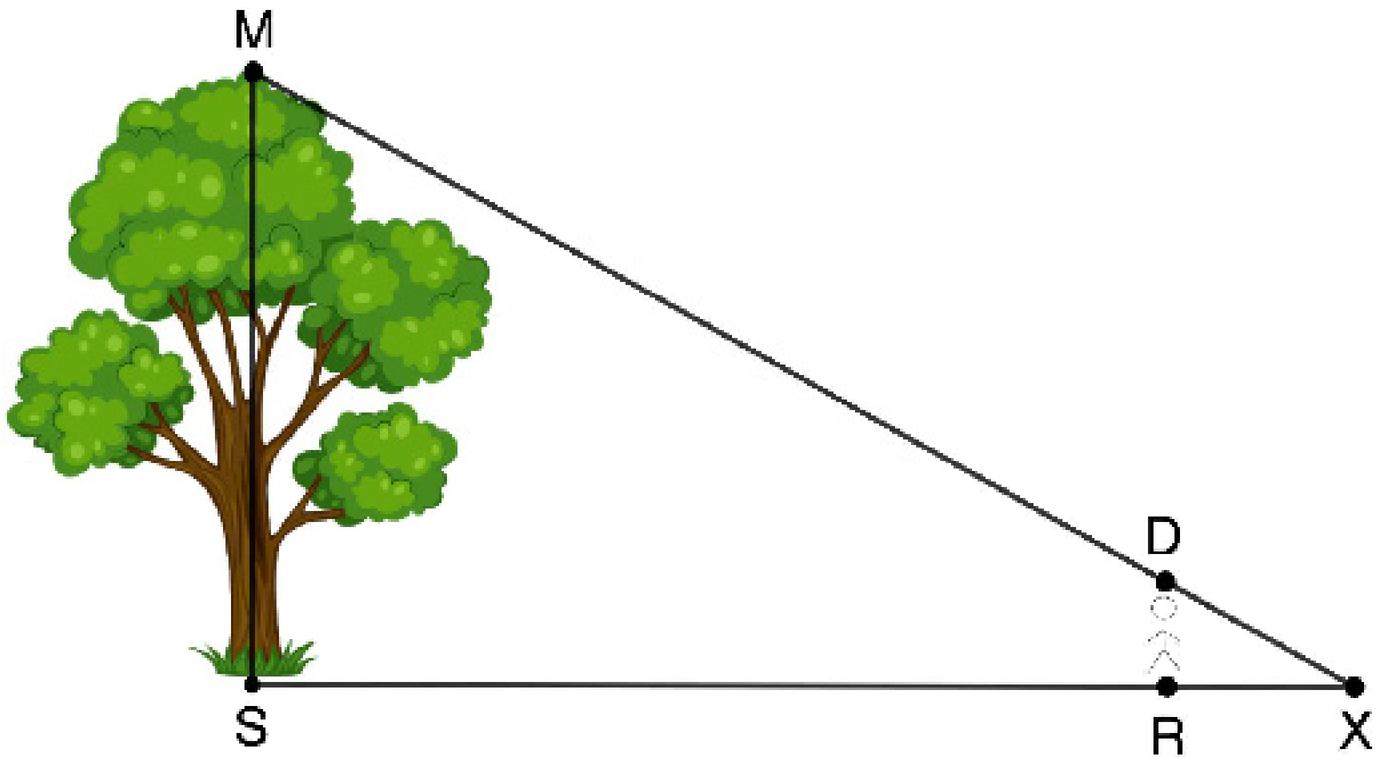
In similar triangles, **corresponding angles** have equal measures and are opposite corresponding sides.

## 6.6.3 Guided Instruction: Using Similar Triangles to Solve Real-World Problems

The largest native tree in Florida is a bald cypress located in Hamilton County. It's 101 feet (ft.) tall. In a local park, there is a very tall tree. Ryan wants to find out if the tree in the local park is close in height to the bald cypress.

When two objects are near one another, the sun hits those objects at approximately the same angle. Ryan walks toward the tree until his shadow and the tree's shadow end at the same point. He is exactly 5.5 ft. tall and casts a shadow of 8 ft. When the shadows are at the same point, Ryan is standing 56 ft. from the base of the tree.

1. Label the diagram with the given information.

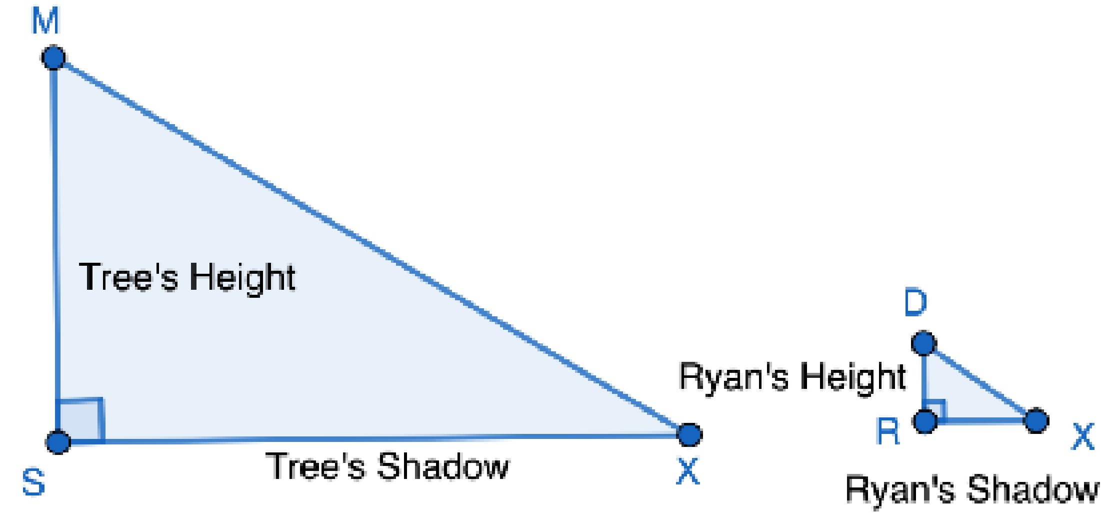



2. How long is the tree's shadow?






3. The diagram shows two similar triangles that can be used to represent the situation.




a. Mark the corresponding angles and label the sides with the known measurements.

b. Determine the constant of proportionality, relating the length of the tree's shadow to the length of Ryan's shadow.

c. Use the constant of proportionality to determine the height of the tree.

4. Explain whether or not the height of the tree in the park is close in height to the bald cypress in Hamilton County.

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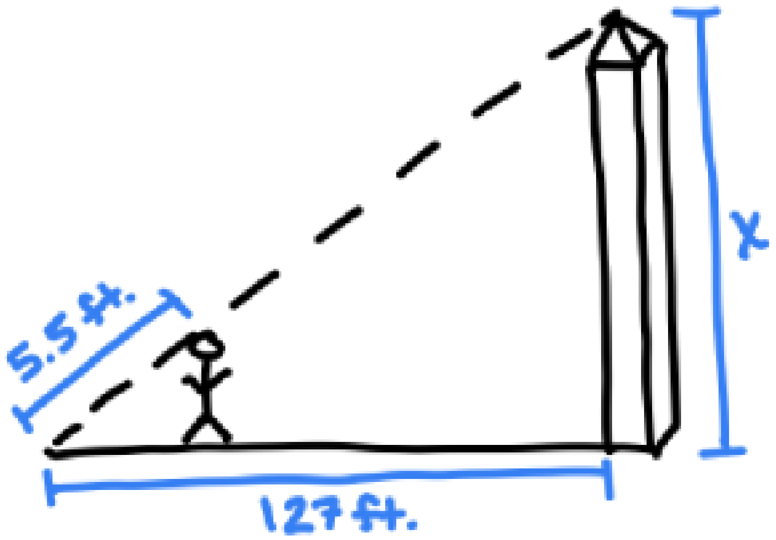
## 6.6.4 Collaboration: Error Analysis

Erika and Michael were working together on the following problem.

A man is standing near the Washington Monument. He is 5.5 feet (ft.) tall and casts a shadow that is 1.25 ft. long. The shadow of the Washington Monument is 127 ft. long. What is the height of the Washington Monument?

1. They drew the following visual representation to help solve the problem.




a. There is an error in their diagram. Describe the mistake they made.

b. Draw a correct diagram. Label the sides with the known measurements.

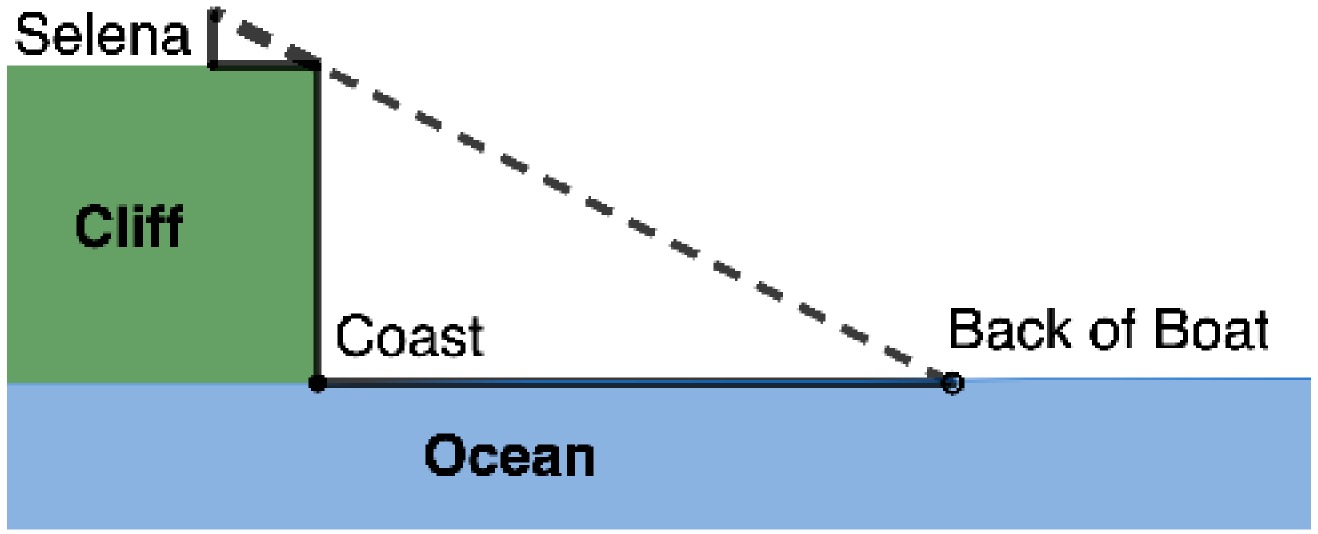
c. Find the height of the Washington Monument.






## 6.6.5 Collaboration: Using Similar Triangles to Solve Real-World Problems

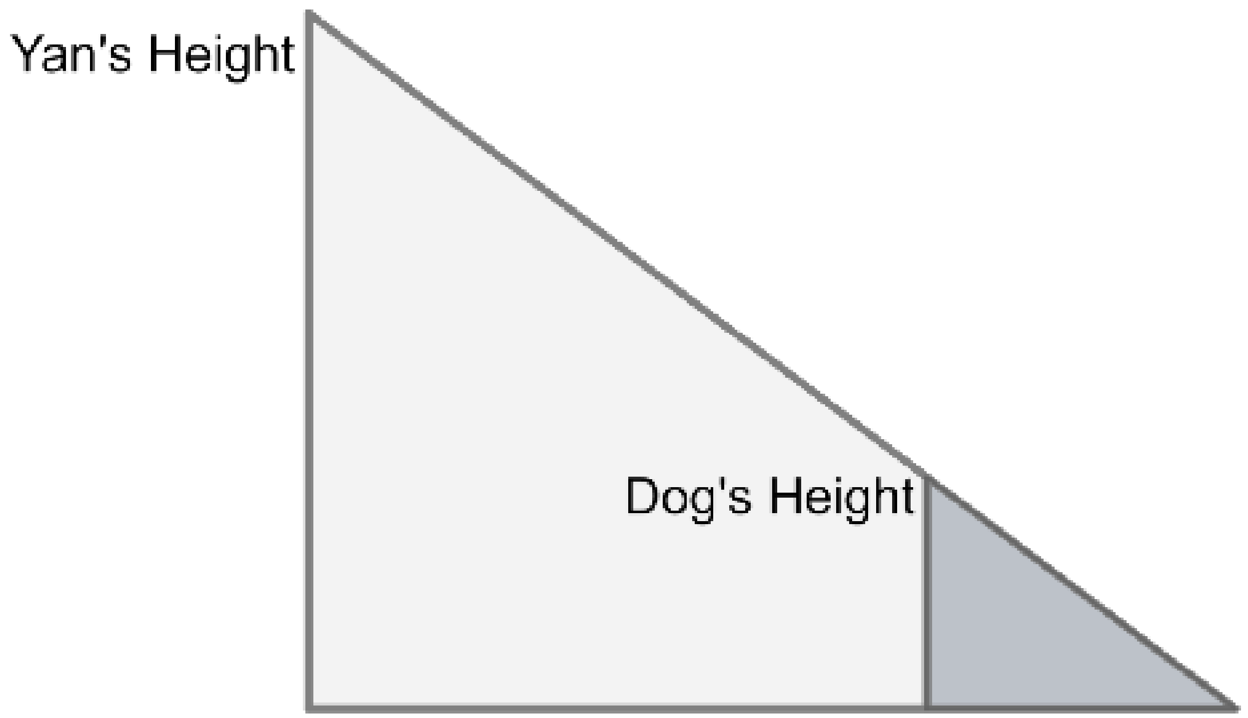
1. While hiking, Selena comes to a cliff overlooking the ocean. She notices a ship in the distance and wonders how far the ship is from the coast. She knows that the cliff is 34.5 feet (ft.) above the water. If she stands 10 ft. from the edge of the cliff, she can visually align the top of the cliff with the water at the back of the ship. Her eye level is $5\frac{3}{4}$ ft. above the ground.




a. Label the similar triangles with the given information.

b. Find the distance from the coast to the ship.

2. Yan is walking his dog. At a certain point in their walk, he notices that their shadows end at the same spot. Yan's dog is walking 24 inches (in.) in front of him. His dog's height is 27 in. and is casting a shadow 18 in. long.




a. Label the similar triangles with the given information.

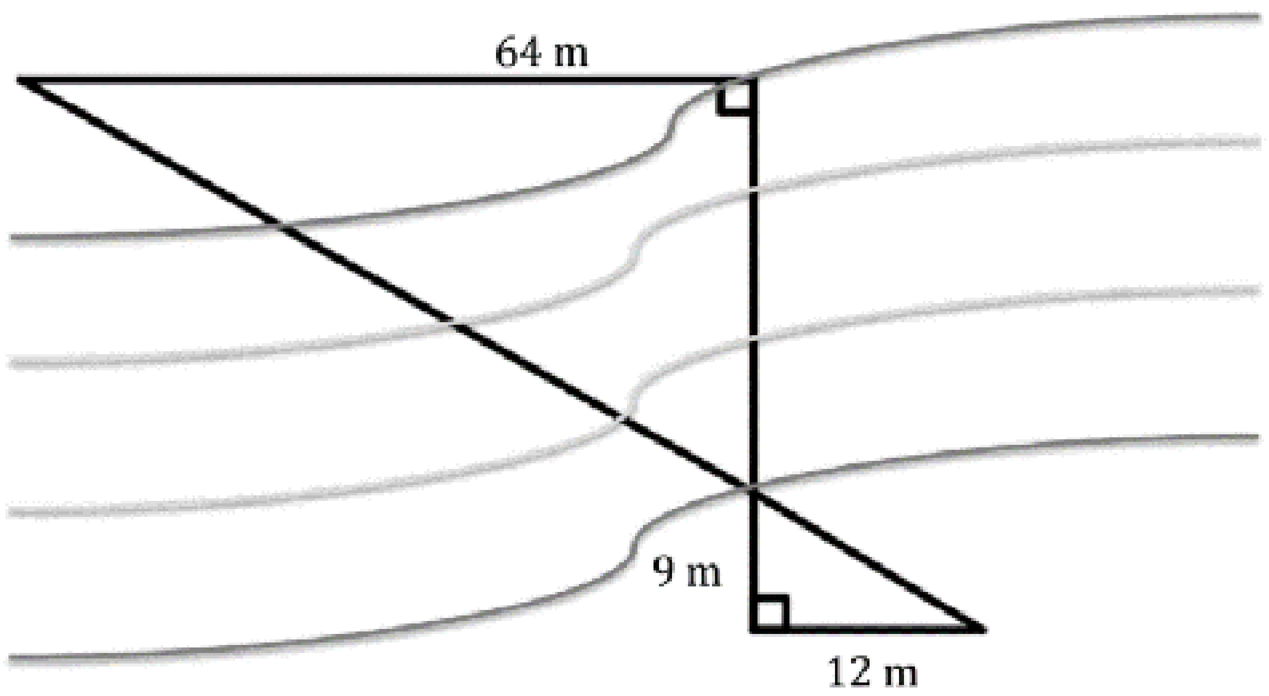
b. Find Yan's height.






## 6.6.6 Your Turn!

1. A surveyor is measuring the width of a river for a future bridge. He is unable to measure across water, so he uses similar triangles and takes a series of measurements on land. The diagram shows his measurements.




Use the surveyor's measurements to determine the width of the river.

## 6.6.7 Wrap-Up: Ticket Out the Door

1. Shanice and her younger brother are walking home from school. Shanice notices that her shadow is 8 feet (ft.) long. Shanice is 5 ft. tall and her brother is 3 ft. tall.

   Determine the length of her brother's shadow.